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ABSTRACT

The synthesis of transfer-free graphene is necessary for expanding its industrial applications. Although the direct synthesis of graphene on the insulating substrate via a metal sacrificial film was reported, the growth mechanism of transfer-free graphene still remains to be studied. Herein, a detailed synthesis model of graphene grown from different carbon sources has been established to help in selecting the growth conditions for high-quality graphene. A detailed discussion on the critical influence of dissolution and the diffusion rate of carbon atoms on the growth process has also been presented. The high decomposition temperature carbon sources promote the formation of high-quality monolayers of graphene. The carbon diffusion rate of the Cu film is significantly higher than that of Ni. This promotes the synthesis of graphene from methane and diamond-like carbon. However, adverse effects are exerted on polymethyl methacrylate. Ion implantation technology and different components of the Ni–Cu alloy were used to understand this growth mechanism. This work could guide the growth conditions of transfer-free, large-scale, and high-quality graphene that can be potentially used for the fabrication of a semiconductor or an insulation substrate in theory. The reported method can generate interest in the field and increase the industrial applications of graphene-based devices that exhibit rough or patterned surfaces.

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I. INTRODUCTION

Graphene has interested scientists as it exhibits extraordinary mechanical strength,¹ high carrier mobility,^{2,3} and excellent transmittance.⁴ These properties make it suitable for use as a critical material for the development of various devices such as supercapacitors,^{5,6} flexible materials,^{7,8} and detectors.⁹ To realize the various industrial applications of graphene, it is important to synthesize large-scale and high-quality graphene. The formation of wrinkles or defects in graphene during the transfer process can influence the performance of graphene-based devices, so it is desirable to synthesize the transfer-free graphene film that can be directly formed on various kinds of substrates and combine well with various substrates.¹⁰ So far, a broadly used strategy to grow transfer-free graphene on different substrates is the metal-sacrificial-assisted

synthesis, which means graphene grown directly on insulating substrates by removing the sacrificial metal films as catalysts using a wet etch step.^{11,12} Copper and nickel are widely used as catalysts due to their lattice structures matching the hexagonal structure of graphene.^{13,14} The layer number, quality, and size of graphene films can be influenced by various growth conditions. However, there is a lack of an in-depth understanding of the growth mechanism, which is the precondition for a rational optimization of growth parameters.

The current research landscape around the growth mechanism was focused on graphene growth on copper and nickel foils.^{15,16} Little is known about the growth process of transfer-free graphene. It is widely believed that the growth of graphene occurs predominantly during the process of annealing in the presence of carbon

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atoms dissolving and diffusing through the catalyst. So, the growth mechanism of transfer-free graphene can be revealed by studying dissolution and the diffusion rate of carbon atoms. On the other hand, recent reports have discussed the characteristic features of graphene derived from different carbon sources that range from gas,¹⁷ liquid¹⁸ to organic solid¹⁹ and amorphous-carbon films.²⁰ But the different characterization of graphene films synthesized from these carbon sources under the same annealing conditions has not been studied yet. Also, carbon diffusion barriers such as Al_2O_3 or Cu were introduced to slow down the diffusion process to improve the quality of graphene synthesized.^{21,22} However, a detailed explanation was not provided. Thus, a systematic growth mechanism needs to be proposed to explain the factors affecting the layer number and the quality of transfer-free graphene.

In this work, we compare the difference between graphene films synthesized from different carbon sources, including methane (CH_4), polymethyl methacrylate (PMMA), and diamond-like carbon (DLC) films. The metal-sacrificial-assisted method was used to grow transfer-free graphene. A rapid annealing treatment method was used to dissolve carbon atoms into the metal catalyst film. Subsequently, the carbon atoms were precipitated between the metal film and the substrate to achieve a self-assembly to form graphene. The prepared graphene films grown from CH_4 , PMMA, and DLC were labeled as M-G, P-G, and D-G, respectively. Various experiments were designed and performed to analyze how different carbon sources and different composition of metal films influenced the quality of graphene. Thus, we explored that the dissolution and diffusion rates of carbon atoms were the major factors influencing graphene growth. The ion implantation technology was used to support our hypothesis. Our study offers new insights into the growth mechanism of transfer-free graphene, defining guidelines for the fabrication of high-quality graphene films.

II. EXPERIMENTAL SECTION

A. Sample preparation

A multilayered substrate of Ni/Cu/ SiO_2 /Si was prepared to achieve graphene growth. The $10 \times 10 \text{ mm}^2$ SiO_2 (300 nm thick)/Si substrate was ultrasonicated for cleaning. Acetone, alcohol, and de-ionized water were used in turn during the cleaning process. Following this, the metal sacrificial layer was deposited onto the SiO_2 /Si substrate by separately evaporating a 25 nm-thick copper film and a 50 nm-thick nickel film (deposition pressure: $\sim 10^{-7}$ Torr; deposition rate: 0.1 nm s^{-1}) using an electron beam evaporation coating system (Detech, DE400DHL). The Ni and Cu ratios can be easily controlled by changing the thickness of the Ni and Cu films.

To obtain M-G, CH_4 (6 SCCM) gas was fed to provide carbon precursors. For P-G samples, PMMA particles were dissolved in a solution of anisole. The PMMA content was 4%. Subsequently, the 4% PMMA solution was spun on the multilayered substrate at a speed of 3000 r/min for 1 min and dried at 100°C for 20 min. For D-G, samples were prepared by depositing a 10 nm DLC film on the multilayered substrate using the magnetron sputtering system with a deposition rate of 0.03 nm s^{-1} .

B. Synthesis of graphene

The growth process was performed in a CVD furnace (Xiamen G-CVD Material Technology Co., Ltd) under atmospheric pressure. Samples were annealed at a temperature of 1000°C for 3 min under the conditions of rapid heating (200 SCCM of Ar and 50 SCCM of H_2 atmosphere). In order to achieve rapid heating, the furnace was placed on the right side of the samples to heat to 1100°C for approximately 25 min, following which the furnace was slid to the left to heat the samples to 1000°C for 3 min. Finally, the samples were cooled at a rate of $\sim 20^\circ\text{C/min}$. The entire growth process could be completed in $\sim 1 \text{ h}$.

C. The etching process for the metal sacrificial layer

First, the organic residue on the surface of samples was removed using acetone and anhydrous ethanol. Following this, all samples were immersed into a solution of nitric acid (HNO_3 : $\text{H}_2\text{O} = 1:4$) for 10 s to remove the graphene synthesized on the surface of the metal film (Fig. S1 in the [supplementary material](#)). The samples were treated with FeCl_3 solution (0.1M) containing a few drops of HCl for $\sim 1 \text{ h}$ to etch the metal film. After the Ni-Cu alloy was completely etched, graphene synthesized on the SiO_2 /Si substrate was obtained, which was then immersed in de-ionized water several times (10 min each time), following which it was air-dried.

D. Characterization of graphene

The fabricated graphene was characterized using the Raman spectroscopy technique (Thermo Fisher DXRxi, laser excitation: 532 nm; 900 grooves/mm gating; spot size: $\sim 1 \mu\text{m}$). The surface morphology of graphene was characterized using the scanning electron microscopy (SEM, Hitachi S-4800) technique. X-ray photoelectron spectroscopy (XPS, Kratos, Axis Ultra), equipped with a monochromatic Al $K\alpha$ x-ray source (1486.7 eV), was used to analyze the hybridization of carbon atoms.

III. RESULTS AND DISCUSSION

Figure 1 illustrates the transfer-free process of synthesizing M-G, P-G, and D-G. Carbon sources and the metal sacrificial layer were fabricated on the SiO_2 /Si substrates. Following this, the thermal treatment method was applied under an atmosphere of Ar/ H_2 (200/50 SCCM). As the temperature increased, copper and nickel films intermixed and formed a gradient of Ni-Cu alloy.²³ When the temperature reached a specific value, the carbon sources decomposed, and the carbon atoms dispersed into the metal film. During the heating process at a constant temperature (at 1000°C) and during the cooling process, the carbon atoms diffused from the upper surface of the metal layer to the lower surface and precipitated to generate self-assembled graphene on both the metal film surface and the substrate. Following the removal of the graphene layer on the upper surface of the sample and the metal sacrificial layer, graphene directly synthesized on the substrate was obtained.

The Raman spectroscopy technique is commonly used to characterize the layer number and quality of graphene.^{24,25} Figure 2(a) compares the Raman spectra of graphene fabricated from different carbon sources under the same thermal treatment conditions. As we

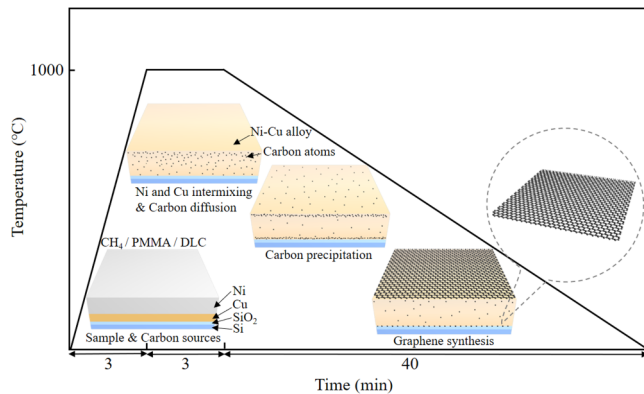


FIG. 1. Schematic illustration of a transfer-free synthesis of graphene from different carbon sources on the SiO_2/Si substrate via a rapid annealing process. The thermal process led to the decomposition of carbon sources and the diffusion of Ni atoms and Cu atoms forming a Ni-Cu alloy. Carbon atoms diffused from the upper surface of the metal layer to the lower surface and self-assembled into graphene on both sides.

know, graphene is characterized by three typical peaks, the D peak ($\sim 1350 \text{ cm}^{-1}$), G peak ($\sim 1580 \text{ cm}^{-1}$), and 2D peak ($\sim 2700 \text{ cm}^{-1}$). The G peak is associated with the doubly degenerate phonon mode, which has its origin in the planar stretching vibrations of the sp^2 hybridized C—C bonds in graphene.²⁶ The D peak, which can be regarded as the defect-induced peak, originates from one iTO phonon and one defect. The D peak and the G peak can be often used to estimate the graphene crystallite size (L_a) by the following equation:²⁷

$$L_a = \frac{560}{E_L^4} \left(\frac{I_D}{I_G} \right)^{-1}, \quad (1)$$

where E_L is the energy of Raman laser excitation (2.33 eV). The 2D peak can be attributed to the two iTO phonons near the K point of the Brillouin zone. The intensity ratio of the 2D band and G band (I_{2D}/I_G) can be used to evaluate the layer number of graphene.

Previous studies have indicated that an I_{2D}/I_G ratio of >2 indicates a monolayer of graphene, while the peak ratio <1 indicates a multilayer of graphene (i.e., layer number more than 10).²⁸ As shown in Fig. 2(a), the D peak was almost non-existent in the Raman spectra of D-G, and I_{2D}/I_G was greater than 2. This indicated that high-quality monolayers of D-G were synthesized. For the M-G sample, the I_D/I_G peak ratio was small, and the I_{2D}/I_G was greater than 1, indicating the presence of few defects and the formation of multilayers. The I_D/I_G peak ratio of P-G was larger than that of M-G, and the I_{2D}/I_G of P-G was less than 1, indicating the presence of a larger number of defects and the formation of multilayers. Moreover, according to the aforementioned equation, the graphene crystallite sizes were approximately 558, 233, and 114 nm for D-G, M-G, and P-G, respectively.

Figures 2(b)–2(e) present the Raman mapping images of D-G and M-G. The images were recorded on an area of $30 \times 30 \mu\text{m}^2$ with a spot size of $2 \mu\text{m}$. The step size was $1 \mu\text{m}$. The analysis of Fig. 2(b) reveals that more than 70% of the area corresponding to D-G exhibited an I_D/I_G ratio <0.2 . The data presented in Fig. 2(c) show that more than 77% of the area had an I_{2D}/I_G ratio >1.2 . Figure 2(d) reveals that 76% of the M-G area was characterized by an I_D/I_G ratio in the range of 0.3–0.6, and Fig. 2(e) shows more than 72% of the area was characterized by an I_{2D}/I_G ratio of >1 . These above results indicated that the quality of D-G was better than that of M-G.

SEM and XPS techniques were used to study graphene coverage and understand the C—C bonding configuration. The experiments were performed with the P-G, M-G, and D-G samples. Figure 3(a) shows the SEM image of graphene synthesized on the upper surface of the P-G sample. The bright curves represent the graphene edges, and the lighter colored areas represented graphene. After disposing of the samples, the surface topographies of P-G and M-G are shown in Figs. 3(b) and 3(c). It was observed that P-G covered the entire substrate with graphene wrinkles. Since the Ni-Cu alloy and graphene exhibited different thermal expansion coefficients, graphene folded as the temperature dropped.²⁹ Also, the coverage of MG was estimated to be $>80\%$. The bright area represented by dot A represented graphene and dot B represented the substrate. Further studies on D-G were conducted using the XPS technique. The primary peak at 284.7 eV corresponded to the sp^2

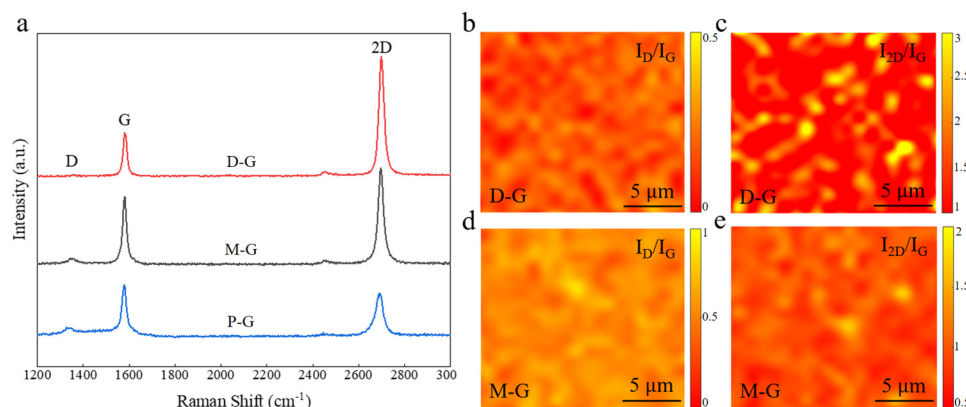


FIG. 2. (a) Raman spectra of D-G, M-G, and P-G, respectively. (b) and (c) Raman mapping images of I_D/I_G and I_{2D}/I_G peak ratios of D-G. (d) and (e) Raman mapping images of I_D/I_G and I_{2D}/I_G peak ratios of M-G.

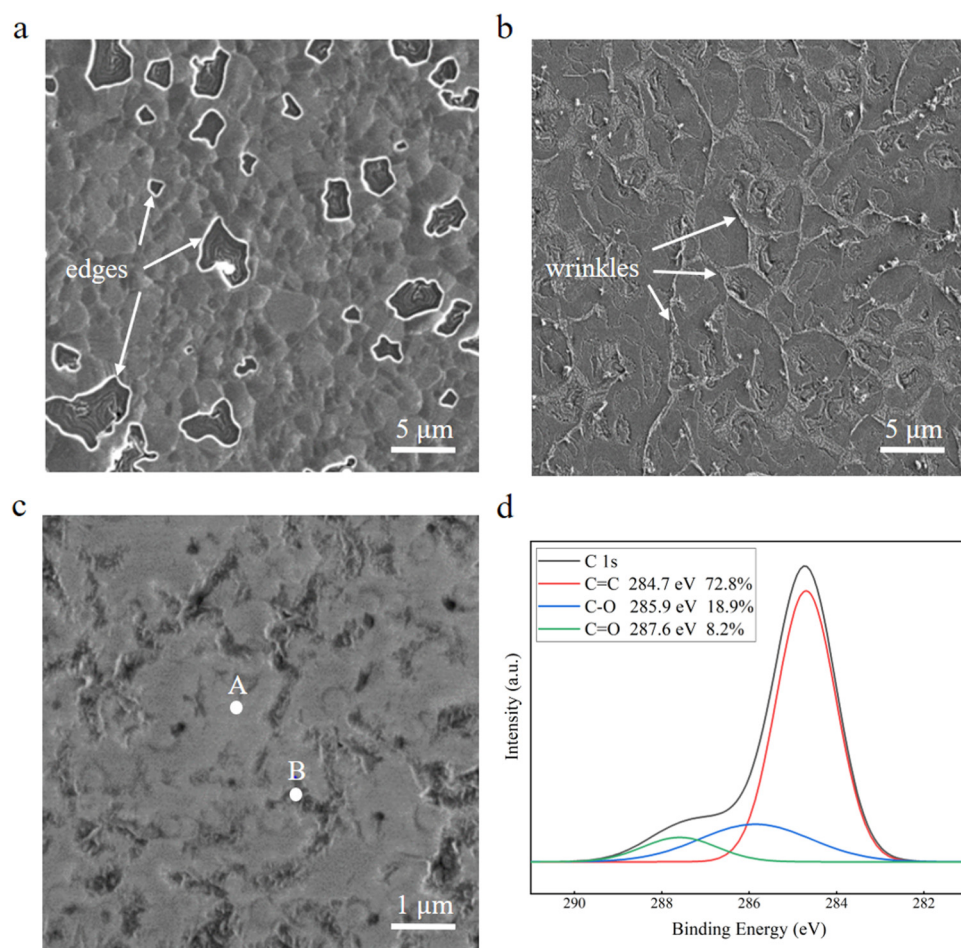


FIG. 3. SEM images of (a) graphene growth on the upper surface of the metal film, (b) P-G, and (c) M-G. (d) XPS C 1s spectrum of D-G. The C 1s peak can be split into three Lorentzian peaks at 284.7, 285.9, and 287.6 eV.

carbon,³⁰ indicating that over 72.8% of the carbon atoms formed C=C bonds. The results were consistent with Raman's results. The small peaks at 285.9 eV and 287.6 eV represented other forms of carbon atoms. As graphene absorbed carbon-containing molecules such as carbon dioxide from the air, the formation of other forms of carbon was inevitable. The XPS C 1s spectrum of M-G has been presented in Fig. S2 in the [supplementary material](#). This method can be used to directly prepare high-quality graphene on any high-temperature-resistant substrate. We studied the similarities and differences in the growth mechanism of graphene derived from different carbon sources to widen the scope of applications. Moreover, the ion implantation technology was used to validate the mechanism.

The layer number and the quality of graphene are sensitive to various growth conditions, such as annealing temperature, annealing time, carbon sources, etc. It is crucial to understand the growth mechanism of graphene to optimize these parameters. It is generally believed that there are four steps in graphene growth: (i) decomposition of a carbon source, (ii) diffusion of carbon atoms through the metal film, (iii) graphene nucleation, and (iv) extension of graphene domains.^{31–34} Figure 4(a) presents the essential

reactions involving hydrogen during the growth of graphene. Decomposition of the carbon source by the reaction of hydrocarbons and hydrogen at elevated temperature is shown in Reaction 1. Carbon atoms diffuse into the metal film, precipitate between the metal and the substrate, and conjugate with the surrounding carbon atoms to form graphene domains. The Gibbs free energy of carbon in the graphene state is lower than that of other states (such as hydrocarbons).³⁵ This is the thermodynamic driving force for the process. The extension of graphene domains is shown in Reaction 2, which is an endothermic reaction. When the temperature exceeds a certain value, carbon radicals bind with graphene domains during diffusion to form larger domains. This certain value is termed as the minimum growth temperature [approximately 700 °C, shown in Fig. S3(a) in the [supplementary material](#)]. As H₂ plays the dual role of a carbon source activator and a graphene etchant, etching of graphene occurs simultaneously (Reaction 3). The mixed effect of extension and etching of graphene domains resulted in the formation of different characteristics of graphene. Also, grain boundaries were formed due to different orientations of graphene domains.^{36,37} So, the lower the density of graphene domains, the better the quality of graphene.

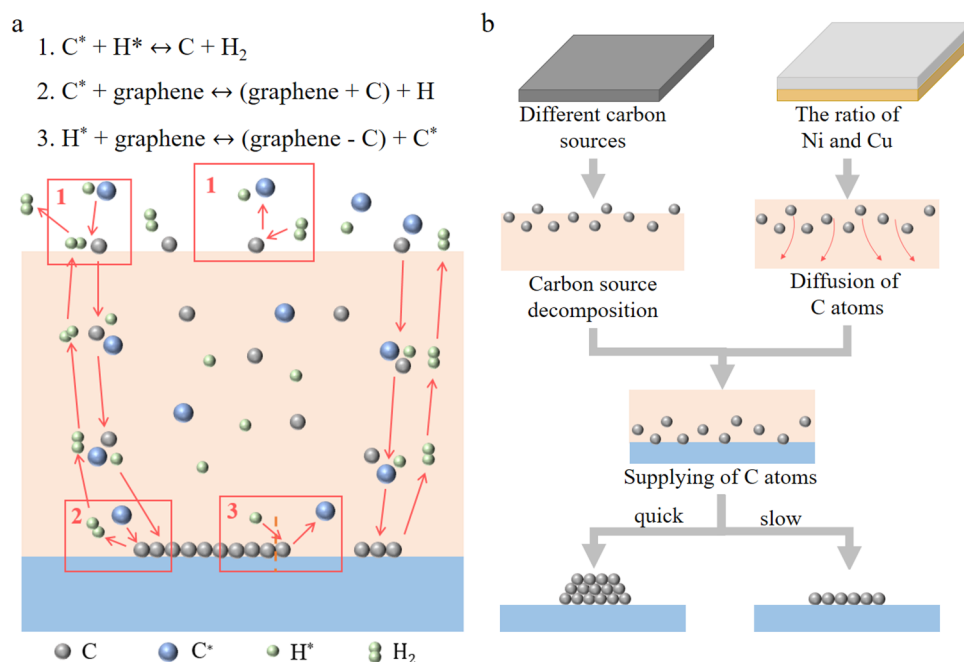


FIG. 4. (a) Basic reactions occurring during graphene growth. C represents the carbon atom, C^* represents the carbon or hydrocarbon radical, and H^* represents the hydrogen radical. (b) Schematic diagram of the mechanism of graphene growth. Carbon source decomposition temperature and carbon diffusion rate affect carbon concentration near the substrate, leading to the synthesis of different types of graphene.

Figure 4(b) shows two important processes (decomposition of carbon source and diffusion of carbon atoms) occurring during the process of graphene growth. The dissolution rate and the diffusion rate of the carbon atoms codetermine the supply rate of carbon atoms near the substrate, which is positively related to the density of graphene domains and the layer number of graphene. According to thermogravimetric measurements, PMMA begins to volatilize and decompose at approximately 200 °C and cracks into carbon radicals completely at approximately 400 °C. In this case, the carbon source decomposes easily, resulting in the oversupply of carbon atoms. This leads to the formation of increased layers and lower quality of graphene. It is difficult to optimize the quality of graphene as the carbon atoms pile up too fast to repair the defects generated during the growth process. As shown in Figs. 5(a) and 5(b), when the annealing temperature lies in the range of 850–1050 °C, the analysis of the Raman spectra of P-G revealed that the I_D/I_G and I_{2D}/I_G (<1) ratio remained almost constant. These results indicated that a defective multilayer of P-G was obtained, and the quality of P-G could not be improved significantly. The results were consistent with the theory.

In the case of methane, the decomposition temperature is generally higher than 1000 °C. Even in the presence of metal catalysis and a suitable hydrogen environment, the temperature exceeds 800 °C,³⁸ leading to a slow dissolution rate of carbon atoms. So the quality of M-G is better than that of P-G, and fewer layers are obtained in theory. Figures 5(a) and 5(b) revealed that at high temperatures, the I_D/I_G ratio of M-G was smaller than that of P-G. The

I_{2D}/I_G ratio of M-G was larger. On the other hand, the density of graphene domains decreases with an increase in temperature. Figure 5(a) shows that as the temperature increased, the I_D/I_G ratio decreased, indicating the formation of higher-quality M-G. The DLC film is composed of sp^2 and sp^3 hybridized carbon. Hence, more energy is required to produce carbon radicals. This results in the lower dissolution rate of carbon atoms on the metal film. Figures 5(a) and 5(b) reveal that a lower I_D/I_G ratio and a higher I_{2D}/I_G ratio can be obtained, and the quality of D-G can be improved by adjusting the annealing temperature. A high-quality monolayer of D-G was synthesized above 950 °C. The carbon sources with high composition temperature promote the formation of high-quality monolayers of graphene.

To verify our explanation, the ion implantation technology was applied to the multilayered substrate of Ni/Cu/SiO₂/Si. Implanted carbon ions served as a limiting case of the carbon source for graphene growth, and no additional carbon source was provided. The process of carbon ion implantation was performed on the multilayered substrate using the dual hollow cathode metal ion source (Duhocamis) test bench at Peking University. The experiments were conducted at room temperature with the ion energy of 20 keV and the fluence of 4×10^{15} ions cm⁻² (corresponding to the equivalent of monolayer graphene). The Stopping and Range of Ions in Matter (SRIM) computer program³⁹ was used to calculate the carbon concentration vs implantation depth [Fig. 5(c)]. Calculations showed that most carbon ions were injected into the nickel film and a few into the copper film. According to the

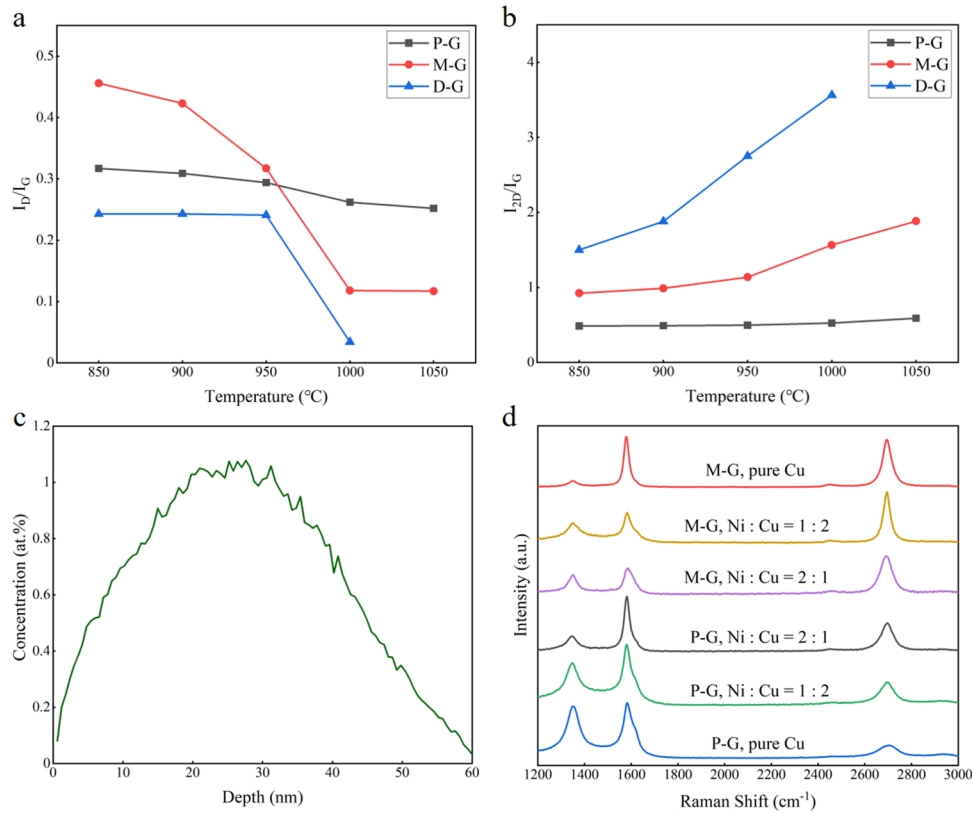


FIG. 5. The (a) I_D/I_G and (b) I_{2D}/I_G peak ratios of P-G, M-G, and D-G grown at different thermal temperatures. (c) Depth profile of carbon concentration distribution of carbon ions (20 keV, 4×10^{15} ions cm^{-2}) into the Ni film. (d) Raman spectra of P-G and M-G at the same thermal treatment with different components of Ni and Cu films.

growth mechanism presented above, a high rate of carbon atoms dissolved into metal films causes the oversupply of carbon atoms near the substrate. This leads to the formation of multilayers and low quality graphene. For the implanted carbon ions, a carbon source decomposition step was absent, and it could be regarded as the limiting case with an infinite dissolution rate for carbon atoms. So the quality of graphene grown from carbon ions should be worse than P-G in theory. Experiments under the same annealing conditions confirmed this suppose. Although the fluence of carbon atoms was 4×10^{15} ions cm^{-2} , which corresponded to a monolayer of graphene, multilayer graphene with a high I_D/I_G ratio was obtained [Fig. S3(b) in the [supplementary material](#)]. The situation resembled that of P-G, and the results revealed that the high dissolution rate of carbon atoms adversely affects graphene growth. Also, the effect of the dissolution rate can be proved by changing the fluence of CH_4 . Theoretically, when the fluence of CH_4 is high enough, the dissolution rate of carbon atoms can be comparable to the case of PMMA, causing M-G to tend to resemble P-G. This phenomenon was observed with 50 SCCM of CH_4 [Fig. S3(b) in the [supplementary material](#)].

On the other hand, the effect of the diffusion rate of carbon atoms in a metal film can be studied by changing the Ni/Cu ratio

of the metal layer. The diffusion coefficient of the carbon atoms (D) in metal is given as follows:⁴⁰

$$D = D_0 e^{-E_D/kT}, \quad (2)$$

where D_0 is the frequency factor that depends on the crystal lattice, E_D is the diffusion activation energy, and k is the Boltzmann constant. The diffusivity of Cu is higher than that of Ni, and this can be primarily attributed to a larger lattice constant and a lower diffusion barrier (as compared to Ni).⁴¹ So, the higher the Ni:Cu ratio, the lower the diffusion rate of the carbon atoms. For P-G, a low decomposition temperature leads to a high dissolution rate of carbon atoms. The rate-limiting step of the process of P-G growth is the diffusion of carbon atoms. A higher diffusion coefficient leads to more supplies of carbon atoms near the substrate and the formation of sp^3 hybridized amorphous carbon. Figure 5(d) shows that P-G did not grow with the sample of pure Cu film at 1000 °C over 3 min. The results supported our explanation. With an increase in the Ni content, the quality of graphene became better under the same annealing conditions. As for M-G, when the content of Ni was high, carbon atoms diffused slowly and the local carbon atoms accumulated near the graphene nucleation sites,

resulting in the formation of multilayers and defects. In conclusion, the decomposition of carbon source and the diffusion of carbon atoms have synergistic effects on the quality and the layer number of graphene by affecting the supply rate of carbon atoms near the substrate.

IV. CONCLUSION

In summary, we have compared the characterization of transfer-free graphene films derived from different carbon sources under the same annealing conditions. By studying the quality and the layer number of P-G, M-G, and D-G, we proposed a growth mechanism of graphene that the dissolution and diffusion rate of carbon atoms influenced the supply rate of carbon atoms near the substrate and affected the quality and thickness of graphene. Moreover, a detailed growth mechanism based on different carbon sources was validated by using ion implantation technology and different components of the Ni-Cu alloy. We proved that for carbon sources with high decomposition temperatures, increasing annealing temperature and higher copper content in metal films could result in better quality of graphene. In this work, the high-quality transfer-free monolayer of graphene was synthesized. The conditions of solid and gaseous carbon sources for better preparation of graphene were theoretically identified. The proposed mechanism can be generalized to guide graphene growth directly on any heat-resistant substrate (such as quartz and sapphire), indicating the feasibility of fabricating graphene-based devices with rough, large-area or patterned surfaces.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for a discussion of the etching process, the additional XPS spectrum of M-G illustrating the hybrid type of carbon atoms, and a more in-depth Raman analysis of the influence of minimum growth temperature and carbon sources.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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